

Intermediate Elective Option 2

Macey Hartmann

2026-05-15

```
# general use
library(tidyverse)

# file organization
library(here)

# color display helper
library(prismatic)

# additional scales and formatting
library(scales)

# font support
library(showtext)

# Google font
font_add_google("Lato", "lato")
showtext_auto()

# read in vegetation transect data
veg <- read_csv(here("data", "veg.csv"))

# read in pool-level metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

1. Explain your inspiration

I chose Tobias Stalder's "Hiking Locations in Washington" circular bar chart as my inspiration. This visualization uses `coord_polar()` in `ggplot2` to arrange grouped bars radially. His

visualization contains bar height, fill color, and an overlaid point and segment.

This format makes sense for the vernal pool vegetation dataset because the data contains multiple discrete pools (analogous to Stalder's hiking regions), each with several continuous summary metrics worth comparing in a single figure. The circular layout compares many pools at once without a cluttered axis and the circular layout feels fitting for a dataset about pools.

My visualization will display the following variables from `veg.csv` and `vp_veg_metadata.csv`:

- **Vernal pool** (`transect_name`): the categorical x-axis, arranged radially
- **Cumulative percent cover** (`sum_pc`): mapped to bar height
- **Number of plant species recorded** (count of distinct `psoc` values per pool): mapped to fill color via a continuous gradient
- **Mean native percent cover** (filtered from `cover_category == "NATIVE COVER"`): mapped to the point and segment overlaid on each bar

2. Plan your figure

3. Code your figure

```
# create new object from veg
pools_vp <- veg |>
  # filter to only vernal pool transects
  filter(str_detect(transect_name, "VP")) |>
  # complete all combinations of year, pool, species, distance
  # fill missing percent cover values with 0
  complete(year, transect_name, psoc, transect_distance,
            fill = list(percent_cover = 0)) |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quadrats
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE),
            .groups = "drop") |>
  # create year_pool column from year and transect name, keep originals
  unite("year_pool", year, transect_name, remove = FALSE) |>
  # join metadata to get number of quadrats per pool-year
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total / number of quadrats
  mutate(mean_pc = sum_pc / num_quad)|>
  # rename transect_name.x back to transect_name after join conflict
```

```

rename(transect_name = transect_name.x) |>
# drop redundant columns brought in by metadata join
select(-year.x, -year.y, -transect_name.y)

pools_vp |>
  group_by(transect_name) |>
  summarise(
    # sum mean percent cover across all years and species per pool
    sum_cover = sum(mean_pc, na.rm = TRUE),
    # mean percent cover across all years and species per pool
    mean_cover = mean(mean_pc, na.rm = TRUE)
  ) -> summary_cover

# count distinct species per pool
pools_vp |>
  group_by(transect_name) |>
  summarise(n = n_distinct(psoc[mean_pc > 0])) -> species_counts

# join species counts back to summary
left_join(summary_cover, species_counts, by = "transect_name") -> summary_all

```

```

# preview color palette in console
prismatic::color(c("#6C5B7B", "#C06C84", "#F67280", "#F8B195"))

```

```

<colors>
#6C5B7BFF #C06C84FF #F67280FF #F8B195FF

```

```

# store figure as object
vp_plot <- ggplot(summary_all) +

  # custom circular grid lines
  geom_hline(yintercept = 0, color = "lightgrey") +
  geom_hline(yintercept = 50, color = "lightgrey") +
  geom_hline(yintercept = 100, color = "lightgrey") +
  geom_hline(yintercept = 150, color = "lightgrey") +

  # bars: x = pool ordered by cumulative cover, y = cumulative cover, fill =
  ↪ species count
  geom_col(aes(
    x = reorder(str_wrap(transect_name, 5), sum_cover),
    y = sum_cover,

```

```

    fill = n),
    position = "dodge2",
    show.legend = TRUE,
    alpha = .9) +

# fill gradient matching Stalder's color palette
scale_fill_gradientn("Number of species",
  colours = c("#6C5B7B", "#C06C84", "#F67280", "#F8B195")) +

# lollipop segment: dashed line from 0 to fixed height for reference
geom_segment(aes(
  x = reorder(str_wrap(transect_name, 5), sum_cover),
  y = 0,
  xend = reorder(str_wrap(transect_name, 5), sum_cover),
  yend = 150),
  linetype = "dashed",
  color = "gray12") +

# lollipop dot: positioned at mean cover value per pool
geom_point(aes(
  x = reorder(str_wrap(transect_name, 5), sum_cover),
  y = mean_cover),
  size = 3,
  color = "gray12") +

# scale y so bars don't start in the center of the polar plot
scale_y_continuous(
  limits = c(-200, 900),
  expand = c(0, 0),
  breaks = c(0, 200, 400, 600, 800)) +

# title, subtitle, caption
labs(
  title = "\nVernal Pool Vegetation",
  subtitle = paste("\nThis visualization shows cumulative percent cover,",
    "number of plant species recorded, and mean percent
    ↪ cover",
    "across eight vernal pools at Coal Oil Point
    ↪ Reserve.\n",
    sep = "\n"),
  caption = "\nData: Coal Oil Point Reserve vernal pool vegetation
    ↪ monitoring\nVisualization inspired by Tobias Stalder ·
    ↪ tobias-stalder.netlify.app") +

```

```

# wrap bars into circular polar layout
coord_polar() +

# minimal theme matching Stalder's style
theme(
  legend.position      = "bottom",
  axis.title           = element_blank(),
  axis.ticks           = element_blank(),
  axis.text.y          = element_blank(),
  axis.text.x          = element_text(color = "gray12", size = 11),
  panel.background    = element_rect(fill = "white", color = "white"),
  panel.grid           = element_blank(),
  panel.grid.major.x   = element_blank(),
  text                 = element_text(color = "gray12", family = "lato"),
  plot.title           = element_text(face = "bold", size = 22, hjust =
    ↪ 0.05),
  plot.subtitle        = element_text(size = 12, hjust = 0.05),
  plot.caption         = element_text(size = 10, hjust = 0.5)) +

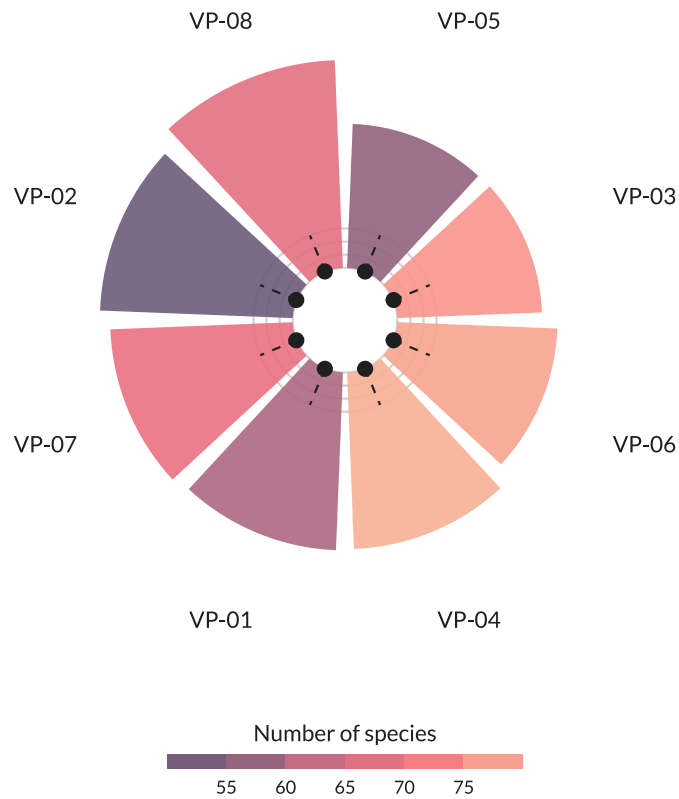
# format the legend color bar
guides(fill = guide_colorsteps(
  barwidth      = 12,
  barheight     = 0.5,
  title.position = "top",
  title.hjust   = 0.5))

# display figure
vp_plot

```

Vernal Pool Vegetation

This visualization shows cumulative percent cover, number of plant species recorded, and mean percent cover across eight vernal pools at Coal Oil Point Reserve.



Data: Coal Oil Point Reserve vernal pool vegetation monitoring
Visualization inspired by Tobias Stalder · tobias-stalder.netlify.app